

AI Model Comparison Cheat Sheet

A quick-reference guide for choosing the right AI model

Snapshot as of July 23, 2026 — this space moves fast; re-check official pricing pages before big decisions

1. Frontier Proprietary Models

Model	Provider	Best For	Context	Price (In/Out per MTok)
Claude Opus 4.8	Anthropic	Complex reasoning, agents, high-stakes coding	200K	\$5 / \$25
Claude Sonnet 5	Anthropic	Balanced daily driver: writing, coding, agents	1M	\$2 / \$10 (intro to Aug 31 '26)
Claude Haiku 4.5	Anthropic	Fast, cheap, high-volume tasks	200K	\$1 / \$5
Claude Fable 5	Anthropic	Anthropic's most capable model overall	1M	\$10 / \$50
GPT-5.6 Sol	OpenAI	Flagship reasoning + coding, cybersecurity	~1M	\$5 / \$30
GPT-5.6 Terra	OpenAI	Balanced everyday work at ~half Sol's cost	~1M	\$2.50 / \$15
GPT-5.6 Luna	OpenAI	Fastest, cheapest tier for high-volume use	~1M	\$1 / \$6
Gemini 3.1 Pro	Google	Deep reasoning, long-document research	1M	\$2 / \$12
Gemini 3.6 Flash	Google	Workhorse: coding, agents, multimodal	1M	\$1.50 / \$7.50
Gemini 3.5 Flash-Lite	Google	High-throughput, low-latency, cheap	1M	\$0.30 / \$2.50
Grok 4.5	xAI (SpaceXAI)	Coding & agentic work, strong cost-efficiency	Large	\$2 / \$6

Note: Claude Sonnet 5 moves to \$3/\$15 after Aug 31, 2026. GPT-5.6 and Gemini 3.1 Pro context windows are advertised up to 1M tokens but some tools (e.g. Codex) cap around 272K. Grok 4.5's exact context window was not published at launch.

2. Open-Weight & Self-Hosted Models

Open-weight models closed much of the gap with closed frontier models in 2026. Good choices when you need to self-host, fine-tune, or avoid sending data to a third-party API.

Model	License	Best For
DeepSeek V4-Pro	MIT	All-round leader on open leaderboards; agentic coding + reasoning
Qwen3.6 / Qwen3-Coder	Apache 2.0	Coding-focused, most permissive license, easy commercial use
Kimi K2.6 / K3	Varies (K3 unlicensed at launch)	Long-horizon agentic and tool-use tasks, sub-agent parallelism
GLM-5.2	MIT	Value pick for coding & agents; strong performance-to-cost
Llama 4 Maverick / Scout	Meta Community License	Native multimodality (Maverick); huge context window (Scout)

Run these locally with tools like Ollama or LM Studio for small variants, or access larger variants via hosted APIs (e.g. OpenRouter, Fireworks, Together AI) when your own hardware can't fit them.

3. Quick Decision Guide — By Task

If you need to...	Reach for...
Chat, draft, or answer everyday questions	Claude Sonnet 5 · GPT-5.6 Terra · Gemini 3.6 Flash
Do deep reasoning, research, or analyze long documents	Claude Opus 4.8 / Fable 5 · Gemini 3.1 Pro · GPT-5.6 Sol
Write and debug code, run coding agents	Claude Sonnet 5 / Opus 4.8 · Grok 4.5 · GPT-5.6 Sol · DeepSeek V4-Pro
Process high volumes cheaply (classification, extraction, support tickets)	Claude Haiku 4.5 · Gemini 3.5 Flash-Lite · GPT-5.6 Luna
Self-host for privacy, cost control, or fine-tuning	DeepSeek V4-Pro · Qwen3.6 · Llama 4 · GLM-5.2
Build agents/automations with tool use	Claude Sonnet 5 / Opus 4.8 · Gemini 3.6 Flash · Grok 4.5
Get real-time info tied to social/X activity	Grok 4.5
Work inside a regulated/enterprise environment (compliance, audit logs)	Claude (Enterprise tier) · GPT-5.6 (Business/Enterprise)

4. Teaching Notes — Why This Sheet Will Go Stale

- Model names and tiers change every few weeks. Teach the **categories** (flagship / balanced / budget, closed vs. open-weight) rather than memorizing specific model names — the pattern outlives any single release.
- Every major lab now ships a 3-tier family (e.g. Claude Opus/Sonnet/Haiku, GPT-5.6 Sol/Terra/Luna, Gemini Pro/Flash/Flash-Lite). Teach students to match tier to task: flagship for hard reasoning, mid-tier for daily work, budget tier for high-volume simple tasks.
- Open-weight models (mostly from Chinese labs: DeepSeek, Moonshot/Kimi, Zhipu/GLM, Alibaba/Qwen) closed much of the performance gap with closed models in 2026 — worth a dedicated lesson on when self-hosting makes sense.
- Pricing drops fast. Treat any dollar figure in this sheet as directional, not final — always point students to the provider's live pricing page before a purchasing decision.
- Suggested cadence: refresh this cheat sheet quarterly, and add a short "what changed" note at the top each time so returning students can see the delta rather than re-reading everything.

Sources: official provider pricing/docs pages and independent trackers (OpenRouter, BenchLM, Artificial Analysis) as of July 22–23, 2026.